Compute

(a) $$\int_0^L \sin\frac{n\pi}{L}x \, \sin\frac{m\pi}{L}x \, dx = \begin{cases} 0, & n \neq m \\ \frac{L}{2}, & n = m \end{cases}$$

$n, m \in \mathbb{Z}^+$

(b) $$\int_0^L \cos\frac{n\pi}{L}x \, \cos\frac{m\pi}{L}x \, dx$$

$n, m \in \mathbb{Z}$

$n, m \geq 0$

$$= \begin{cases} L, & n = m = 0 \\ \frac{L}{2}, & n = m > 0 \\ 0, & n \neq m \end{cases}$$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

(c) $\displaystyle\int_{-L}^{L} \cos\frac{n\pi}{L}x \, \sin\frac{m\pi}{L}x \, dx$ $\qquad n \geq 0$

$m > 0$

$n, m \in \mathbb{Z}$

$= 0$

$$= \frac{1}{2}\int_{-L}^{L}\left[\sin\frac{(m+n)\pi}{L}x + \sin\frac{(m-n)\pi}{L}x\right]dx$$

$$= -\frac{1}{2}\,\frac{L}{(m+n)\pi}\cos\frac{(m+n)\pi}{L}x\Big|_{-L}^{L} + \frac{1}{2}\int_{-L}^{L}\sin\frac{(m-n)\pi}{L}x\,dx$$

$$= \frac{1}{2}\int_{-L}^{L}\sin\frac{(m-n)\pi}{L}x\,dx$$

Case 1: $m = n$ $\qquad I = 0$

Case 2: $m \neq n$ $\qquad I = -\frac{1}{2}\,\frac{L}{(m-n)\pi}\cos\frac{(m-n)\pi}{L}x\Big|_{-L}^{L} = 0$

(a) $$\underline{I} = \int_0^L \sin\frac{n\pi}{L}x \sin\frac{m\pi}{L}x\,dx$$

$$= \frac{1}{2}\int_0^L \left[\cos\left(\frac{n\pi}{L}x - \frac{m\pi}{L}x\right) - \cos\left(\frac{n\pi}{L}x + \frac{m\pi}{L}x\right)\right]dx$$

$$= \frac{1}{2}\int_0^L \left[\cos\frac{n-m}{L}\pi x - \cos\frac{n+m}{L}\pi x\right]dx$$

$$= \frac{1}{2}\int_0^L \cos\frac{n-m}{L}\pi x\,dx - \frac{1}{2}\int_0^L \cos\frac{n+m}{L}\pi x\,dx$$

If $n \neq m$ $$= \frac{1}{2}\frac{L}{(n-m)\pi}\sin\frac{n-m}{L}\pi x\Big|_0^L - \frac{1}{2}\frac{L}{(n+m)\pi}\sin\frac{n+m}{L}\pi x\Big|_0^L = 0$$

If $n = m$ $$\underline{I} = \frac{1}{2}\int_0^L 1\,dx - \frac{1}{2}\frac{L}{(n+m)\pi}\sin\frac{n+m}{L}\pi x\Big|_0^L$$

$$= \frac{1}{2}L$$

$$\sin\alpha \sin\beta = \frac{1}{2}\left[\cos(\alpha-\beta) - \cos(\alpha+\beta)\right]$$

$$\cos\alpha \cos\beta = \frac{1}{2}\left[\cos(\alpha+\beta) + \cos(\alpha-\beta)\right]$$

$$\sin\alpha \cos\beta = \frac{1}{2}\left[\sin(\alpha+\beta) + \sin(\alpha-\beta)\right]$$

(b) $I = \int_0^L \cos\frac{n\pi}{L}x \cos\frac{m\pi}{L}x\,dx \qquad n, m \geq 0$

$$= \frac{1}{2}\int_0^L \left[\cos\left(\frac{n\pi}{L}+\frac{m\pi}{L}\right)x + \cos\left(\frac{n\pi}{L}-\frac{m\pi}{L}\right)x\right]dx$$

$$= \frac{1}{2}\int_0^L \cos\frac{n+m}{L}\pi x\,dx + \frac{1}{2}\int_0^L \cos\frac{n-m}{L}\pi x\,dx$$

Case 1: $n = m = 0 \qquad I = L$

Case 2: $n \neq m \quad (n+m>0,\ n-m \neq 0)$

$$I = \frac{1}{2}\frac{L}{(n+m)\pi}\sin\frac{n+m}{L}\pi x\Big|_0^L + \frac{1}{2}\frac{L}{(n-m)\pi}\sin\frac{n-m}{L}\pi x\Big|_0^L$$

$$= 0.$$

Case 3: $n = m \neq 0$

$$I = \frac{1}{2}\int_0^L \cos\frac{2n\pi}{L}x\,dx + \frac{1}{2}\int_0^L 1\,dx$$

$$= \frac{1}{2}\,\frac{L}{2n\pi}\sin\frac{2n\pi}{L}x\Big|_0^L + \frac{1}{2}L$$

$$= \frac{1}{2}L$$

MH4110 Partial Differential Equations

Chapter 5 - Fourier Series

Outline

The goal of this chapter is to find the coefficients in a Fourier series.

- Fourier sine series
- Fourier cosine series
- Full Fourier series
- Convergence of Fourier series

Dirichlet Boundary Conditions – Heat equation

$$LHS = \int_0^L \phi(x) \sin\frac{m\pi x}{L}\,dx$$

Theorem

The heat equation

$$\begin{aligned} u_t &= k u_{xx}, \quad 0 < x < l,\ t > 0, \\ u(0,t) &= u(l,t) = 0, \quad t > 0, \\ u(x,0) &= \phi(x), \quad 0 < x < l, \end{aligned} \tag{1}$$

has the series solution:

$$RHS = \int_0^L \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{L} \sin\frac{m\pi x}{L}\,dx$$

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp\left(-\left(\frac{n\pi}{l}\right)^2 kt\right) \sin\frac{n\pi x}{l}, \tag{2}$$

$$= \sum_{n=1}^{\infty} \int_0^L A_n \sin\frac{n\pi x}{L} \sin\frac{m\pi x}{L}\,dx$$

provided that ϕ has the sine Fourier expansion

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{l}. \tag{3}$$

$$= A_m \frac{L}{2}$$

$$LHS = RHS \Rightarrow A_m = \frac{2}{L}\int_0^L \phi(x) \sin m\pi x/L\,dx$$

Dirichlet Boundary Conditions – Wave equation

Theorem

The wave equation

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l,\ t > 0, \\ &u(0,t) = u(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l, \end{aligned} \tag{4}$$

has the following series solution:

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \sin \frac{n\pi x}{l}, \tag{5}$$

as long as the initial data can be expanded into sine Fourier series

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}, \quad \psi(x) = \sum_{n=1}^{\infty} \frac{n\pi c}{l} B_n \sin \frac{n\pi x}{l}. \tag{6}$$

Neumann Boundary Conditions – Heat equation

Theorem

The heat equation

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \ t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{7}$$

has the series solution:

$$u(x,t) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \cos\frac{n\pi x}{l}, \tag{8}$$

provided that ϕ has the cosine Fourier expansion

$$\phi(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l}. \tag{9}$$

Neumann Boundary Conditions – Wave equation

Theorem

The wave equation

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has the following series solution:

$$u(x,t) = \frac{A_0}{2} + \frac{B_0}{2} t + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l} \right) \cos \frac{n\pi x}{l}, \tag{11}$$

as long as the initial data can be expanded into cosine Fourier series

$$\phi(x) = \frac{1}{2} A_0 + \sum_n A_n \cos \frac{n\pi x}{l}, \quad \psi(x) = \frac{1}{2} B_0 + \sum_n \frac{n\pi c}{l} B_n \cos \frac{n\pi x}{l}. \tag{12}$$

§5.1 The coefficients: Fourier sine series

Motivation: We have found Fourier series of several types, and the Fourier coefficients need to be determined by the initial condition(s).

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Fourier sine series:

We revisit the series

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}, \quad 0 < x < l. \tag{13}$$

It is a series of trigonometric sines, termed as a Fourier sine series. To find the coefficients A_n, let us first study the properties of the sines

$$\left\{ \sin \frac{\pi x}{l}, \sin \frac{2\pi x}{l}, \cdots, \sin \frac{n\pi x}{l}, \cdots \right\}. \tag{14}$$

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- Recall that they are eigenfunctions of the eigenproblem:

$$-X''(x) = \lambda X(x), \quad X(0) = X(l) = 0.$$

§5.1 The coefficients: Fourier sine series

- They are periodic functions. A function ϕ that is defined for $-\infty < x < \infty$ is called periodic if there is a number $p > 0$ such that $\phi(x+p) = \phi(x)$ for all x. A number p for which this is true is called a **period** of $\phi(x)$. For instance, $\sin \omega x$ is $\frac{2\pi}{\omega}$-periodic. The period of $\sin \frac{n\pi x}{l}$ is $\frac{2l}{n}$.

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- For each n, $\sin \frac{n\pi x}{l}$ is an odd function of x for all $x \in (-\infty, \infty)$.

§5.1 The coefficients: Fourier sine series

$\sin \frac{n\pi x}{L}$ period: $\frac{2\pi}{\frac{n\pi}{L}} = \frac{2L}{n}$

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- For each n, $\sin \frac{n\pi x}{l}$ is an odd function of x for all $x \in (-\infty, \infty)$.
- They are mutually orthogonal, namely,

$$\int_0^l \sin \frac{n\pi x}{l} \sin \frac{m\pi x}{l} dx = \begin{cases} l/2, & \text{if } n = m, \\ 0, & \text{if } n \neq m. \end{cases} \tag{15}$$

§5.1 The coefficients: Fourier sine series

- They are periodic functions. A function ϕ that is defined for $-\infty < x < \infty$ is called periodic if there is a number $p > 0$ such that $\phi(x+p) = \phi(x)$ for all x. A number p for which this is true is called a **period** of $\phi(x)$. For instance, $\sin \omega x$ is $\frac{2\pi}{\omega}$-periodic. The period of $\sin \frac{n\pi x}{l}$ is $\frac{2l}{n}$.
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- They are linearly independent.

§5.1 The coefficients: Fourier sine series

- Recall the trigonometric identity:

$$\sin a \sin b = \frac{\cos(a-b) - \cos(a+b)}{2}.$$

We find that for $m, n \geq 1$,

$$\int_0^l \sin\frac{n\pi x}{l} \sin\frac{m\pi x}{l} dx = \frac{1}{2}\int_0^l \cos\frac{(m-n)\pi x}{l} dx - \frac{1}{2}\int_0^l \cos\frac{(m+n)\pi x}{l} dx.$$

It is clear that if $m = n$, the integral value is $l/2$, and it is zero when $m \neq n$.

§5.1 The coefficients: Fourier sine series

- We now determine the coefficients. Multiplying both sides of (13) by $\sin \frac{m\pi x}{l}$ and integrating the resulting equation over the interval $(0, l)$, we find from the orthogonality (15) that

$$\int_0^l \phi(x) \sin \frac{m\pi x}{l} dx = \sum_{n=1}^{\infty} A_n \int_0^l \sin \frac{n\pi x}{l} \sin \frac{m\pi x}{l} dx = A_m \frac{l}{2}.$$

This implies

$$A_m = \frac{2}{l} \int_0^l \phi(x) \sin \frac{m\pi x}{l} dx, \quad m = 1, 2, \cdots. \tag{16}$$

Therefore, **if** $\phi(x)$ **has an expansion** (13)**, then the coefficient must be given by** (16)**.**

§5.1 The coefficients: Fourier sine series

Theorem 1: Fourier sine series-Complete solution formula

Consider the heat equation:

IBVP

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \quad t > 0, \\ &u(0,t) = u(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{17}$$

where k, l are positive constants and ϕ is a given function. It has the solution

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \sin\frac{n\pi x}{l}, \tag{18}$$

where

$$A_n = \frac{2}{l}\int_0^l \phi(x)\sin\frac{n\pi x}{l}dx, \quad n = 1, 2, \cdots. \tag{19}$$

§5.1 The coefficients: Fourier sine series

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where

$$A_n = \frac{2}{l}\int_0^l \phi(x)\sin\frac{n\pi x}{l}dx, \quad n = 1, 2, \cdots. \tag{19}$$

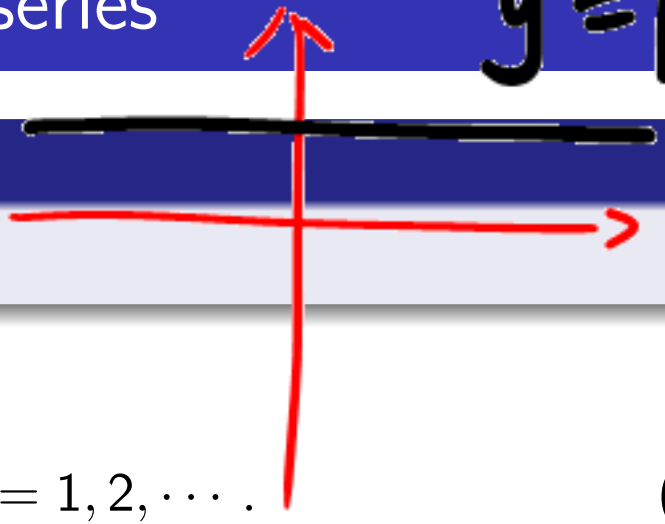
Remark: These are the only possible coefficients in (18). However, whether (18) is in fact valid with these values of the coefficients is a question of convergence.

§5.1 The coefficients: Fourier sine series

Example 1

Expand $\phi(x) = 1, x \in (0, l)$ in Fourier sine series.

$y=1$



Example 1

Expand $\phi(x) = 1, x \in (0, l)$ in Fourier sine series.

Solution: We need to evaluate

$$A_n = \frac{2}{l}\int_0^l 1 \cdot \sin\frac{n\pi x}{l}dx, \quad n = 1, 2, \cdots. \tag{20}$$

More precisely,

$k=1, 2, \cdots$

$$A_n = \frac{2}{l}\int_0^l \sin\frac{n\pi x}{l}dx = -\frac{2}{n\pi}\int_0^l d\cos\frac{n\pi x}{l}$$
$$= -\frac{2}{n\pi}\cos\frac{n\pi x}{l}\Big|_0^l = \frac{2}{n\pi}\left[1-(-1)^n\right]. = \begin{cases} 0, & n=2k \\ \frac{4}{(2k-1)\pi}, & n=2k-1 \end{cases}$$

Therefore, we have the Fourier sine expansion in $(0, l)$:

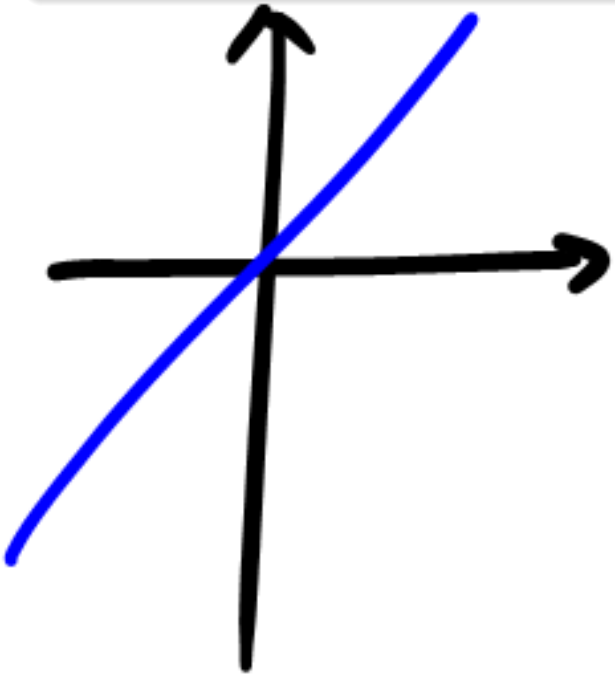
$$1 = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{[1-(-1)^n]}{n}\sin\frac{n\pi x}{l} = \frac{4}{\pi}\left(\sin\frac{\pi x}{l} + \frac{1}{3}\sin\frac{3\pi x}{l} + \frac{1}{5}\sin\frac{5\pi x}{l}\cdots\right).$$

$$1 = \sum_{k=1}^{\infty}\frac{4}{(2k-1)\pi}\sin\frac{(2k-1)\pi x}{L} \tag{21}$$

§5.1 The coefficients: Fourier sine series

Example 3

Expand $\phi(x) = x, x \in (0, l)$ in Fourier sine series and sketch the graph.



Example 3

Expand $\phi(x) = x, x \in (0, l)$ in Fourier sine series and sketch the graph.

Solution: We need to evaluate

$$A_n = \frac{2}{l}\int_0^l x \sin\frac{n\pi x}{l}dx, \quad n = 1, 2, \cdots, \tag{22}$$

by integration by parts. More precisely,

$$\begin{aligned} A_n &= \frac{2}{l}\int_0^l x \sin\frac{n\pi x}{l}dx = -\frac{2}{n\pi}\int_0^l x\, d\cos\frac{n\pi x}{l} \\ &= -\frac{2}{n\pi}x\cos\frac{n\pi x}{l}\Big|_0^l + \frac{2}{n\pi}\int_0^l \cos\frac{n\pi x}{l}dx = \frac{2l}{n\pi}(-1)^{n+1}. \end{aligned}$$

Therefore, we have the Fourier sine expansion in $(0, l)$:

$$x = \frac{2l}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{l} = \frac{2l}{\pi}\Big(\sin\frac{\pi x}{l} - \frac{1}{2}\sin\frac{2\pi x}{l} + \frac{1}{3}\sin\frac{3\pi x}{l}\cdots\Big). \tag{23}$$

§5.1 The coefficients: Fourier sine series

We plot the function $\phi(x) = x$ in $0 < x < l = 3$, and partial sum $S_m(x)$ of the first m terms of the Fourier sine series (29).

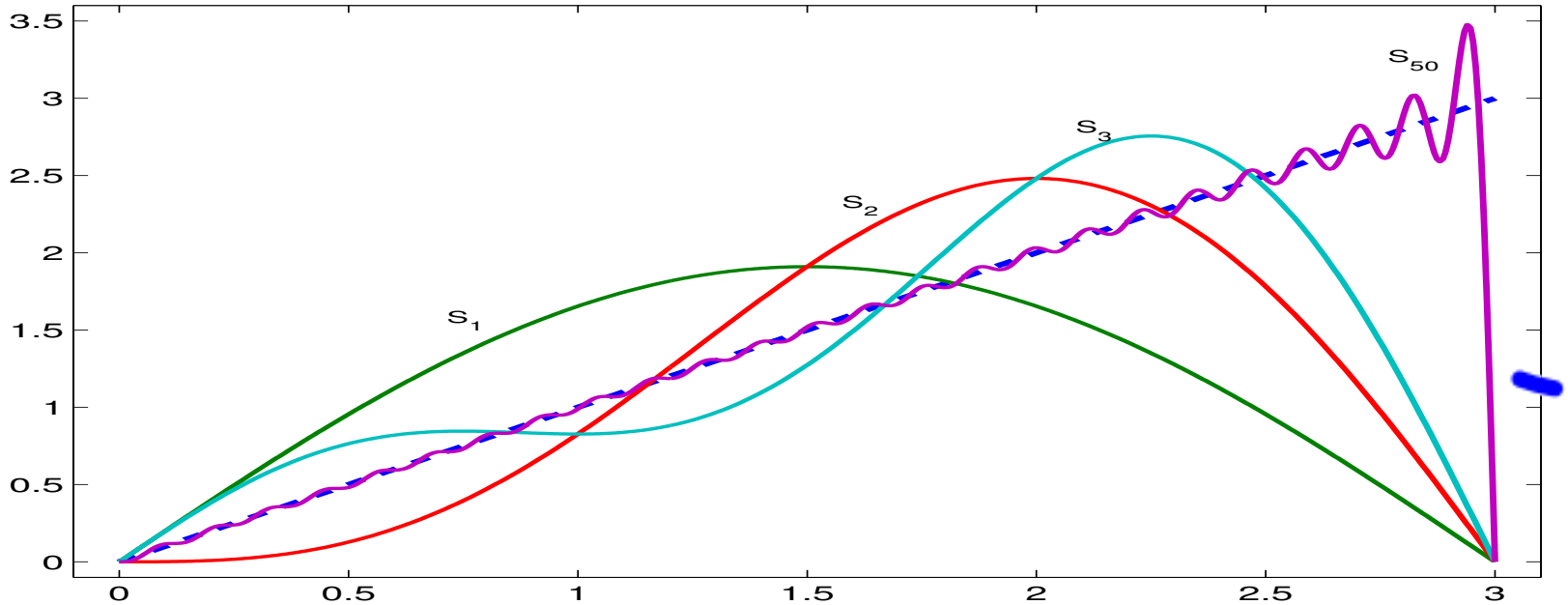


Figure: Fourier sine series of x (marked by dash line) and S_1, S_2, S_3, S_{50}.

$$S_m(x) = \sum_{n=1}^{m} A_n \sin \frac{n\pi x}{L}$$

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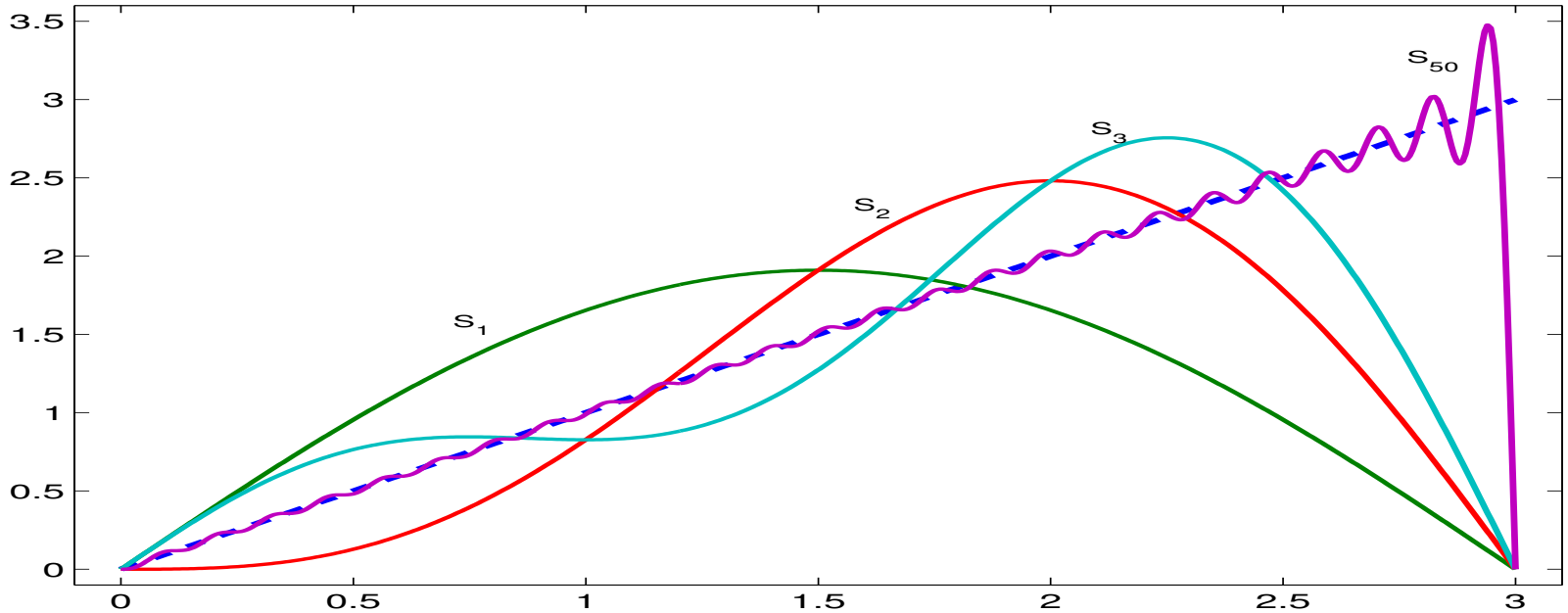

Figure: Fourier sine series of x (marked by dash line) and S_1, S_2, S_3, S_{50}.

We observe the oscillation occurs at the right endpoint (in general, at discontinuity), which is known as the **Gibb's phenomenon.**

§5.1 The coefficients: Fourier cosine series

In the study of wave and heat equations with Neumann boundary conditions in $(0, l)$, we have to consider the series of the form

$$\phi(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{l}, \tag{24}$$

which is referred to as a Fourier cosine series of $\phi(x)$ in $(0, l)$.

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The system

$$\left\{1, \cos \frac{\pi x}{l}, \cos \frac{2\pi x}{l}, \cdots, \cos \frac{n\pi x}{l}, \cdots\right\}, \tag{25}$$

shares similar properties as the sines we discussed previously, but we notice that the cosines are even functions.

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shares similar properties as the sines we discussed previously, but we notice that the cosines are even functions.

Moreover, we have the **orthogonality**:

$$\begin{aligned} &\int_0^l \cos \frac{n\pi x}{l} \cos \frac{m\pi x}{l} dx = 0, \quad \text{if} \quad m \neq n; \\ &\int_0^l 1\, dx = l; \quad \int_0^l \left(\cos \frac{n\pi x}{l}\right)^2 dx = \frac{l}{2}, \quad n = 1, 2, \cdots. \end{aligned} \tag{26}$$

$m = n = 0$

$=$

$\& \; m = n > 0$

§5.1 The coefficients: Fourier cosine series

If $m>0$ $\quad RHS = A_m \int_0^L \left(\cos \frac{m\pi x}{L}\right)^2 dx = A_m \frac{L}{2}$

Thanks to the orthogonality, multiplying both sides of (24) by $\cos \frac{m\pi x}{l}$ and integrating the resulted identity over $(0, l)$, we find that

$$A_m = \frac{2}{L}\int_0^L \phi(x)\cos\frac{m\pi x}{L}\,dx$$

$$\int_0^l \phi(x)\cos\frac{m\pi x}{l}dx = \frac{A_0}{2}\int_0^l 1\cdot\cos\frac{m\pi x}{l}dx + \sum_{n=1}^{\infty} A_n \int_0^l \cos\frac{n\pi x}{l}\cos\frac{m\pi x}{l}dx, \tag{27}$$

If $m=0$

$$RHS = \frac{A_0}{2}\cdot L$$

$$LHS = \int_0^L \phi(x)\,dx$$

$$A_0 = \frac{2}{L}\int_0^L \phi(x)\cos\frac{m\pi x}{L}dx$$

§5.1 The coefficients: Fourier cosine series

Thanks to the orthogonality, multiplying both sides of (24) by $\cos\frac{m\pi x}{l}$ and integrating the resulted identity over $(0, l)$, we find that

$$\begin{aligned}\int_0^l \phi(x)\cos\frac{m\pi x}{l}dx = \frac{A_0}{2}\int_0^l \cos\frac{m\pi x}{l}dx \\ + \sum_{n=1}^{\infty} A_n \int_0^l \cos\frac{n\pi x}{l}\cos\frac{m\pi x}{l}dx,\end{aligned} \tag{27}$$

which implies

$$A_m = \frac{2}{l}\int_0^l \phi(x)\cos\frac{m\pi x}{l}dx, \quad m = 0, 1, 2, \cdots. \tag{28}$$

§5.1 The coefficients: Fourier cosine series

Thanks to the orthogonality, multiplying both sides of (24) by $\cos\frac{m\pi x}{l}$ and integrating the resulted identity over $(0, l)$, we find that

$$\begin{aligned}\int_0^l \phi(x)\cos\frac{m\pi x}{l}dx = \frac{A_0}{2}\int_0^l \cos\frac{m\pi x}{l}dx \\ + \sum_{n=1}^{\infty} A_n \int_0^l \cos\frac{n\pi x}{l}\cos\frac{m\pi x}{l}dx,\end{aligned} \tag{27}$$

which implies

$$A_m = \frac{2}{l}\int_0^l \phi(x)\cos\frac{m\pi x}{l}dx, \quad m = 0, 1, 2, \cdots. \tag{28}$$

At this point, we remark that to get a uniform expression (28) for all the coefficients we put the $\frac{1}{2}$ in front of the constant term in (24).

§5.1 The coefficients: Fourier cosine series

Theorem 2: Fourier cosine series-Complete solution formula

Consider the heat equation:

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$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \quad t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{29}$$

where k, l are positive constants and ϕ is a given function. It has the solution

$$u(x,t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \cos\frac{n\pi x}{l}, \tag{30}$$

where

$$A_n = \frac{2}{l}\int_0^l \phi(x)\cos\frac{n\pi x}{l}\,dx, \quad n = 0, 1, 2, \cdots. \tag{31}$$

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Remark: These are the only possible coefficients in (30). However, whether (30) is in fact valid with these values of the coefficients is a question of convergence.

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Example 2

Expand $\phi(x) = 1, x \in (0, l)$ in Fourier cosine series.

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Solution: We need to evaluate

$$A_n = \frac{2}{l}\int_0^l \cos\frac{n\pi x}{l}dx, \quad n = 0, 1, 2, \cdots.$$

When $n \neq 0$,

$$A_n = \frac{2}{l}\int_0^l \cos\frac{n\pi x}{l}dx = \frac{2}{n\pi}\sin\frac{n\pi x}{l}\Big|_0^l = \frac{2}{n\pi}[\sin n\pi - \sin 0] = 0.$$

When $n = 0$,

$$A_0 = \frac{2}{l}\int_0^l \cos\frac{n\pi x}{l}dx = 2.$$

Therefore, we have the trivial Fourier cosine expansion in $(0, l)$:

$$1 = 1 + 0\cos\frac{\pi x}{l} + 0\cos\frac{2\pi x}{l} + \cdots.$$

Example 4

Expand $\phi(x) = x, x \in (0, l)$ in Fourier cosine series and sketch the graph.

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For $n > 0$, we derive from integration by parts that

$$\begin{aligned} A_n &= \frac{2}{l}\int_0^l x\cos\frac{n\pi x}{l}dx = \frac{2}{n\pi}\int_0^l x\, d\sin\frac{n\pi x}{l} \\ &= \frac{2}{n\pi}x\sin\frac{n\pi x}{l}\Big|_0^l - \frac{2}{n\pi}\int_0^l \sin\frac{n\pi x}{l}dx \\ &= \frac{2l}{n^2\pi^2}\cos\frac{n\pi x}{l}\Big|_0^l = \frac{2l}{n^2\pi^2}\left((-1)^n - 1\right). = \begin{cases} -\dfrac{4L}{(2k-1)^2\pi^2}, & n = 2k-1 \\ 0, & n = 2k \end{cases} \end{aligned}$$

$k = 1, 2, \cdots$

§5.1 The coefficients: Fourier cosine series

Therefore, we have

$$A_{2k} = 0, \quad k = 1, 2, \cdots ; \quad A_{2k+1} = -\frac{4l}{(2k+1)^2\pi^2}, \quad k = 0, 1, 2, \cdots .$$

§5.1 The coefficients: Fourier cosine series

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$$A_{2k} = 0, \quad k = 1, 2, \cdots; \quad A_{2k+1} = -\frac{4l}{(2k+1)^2\pi^2}, \quad k = 0, 1, 2, \cdots.$$

Finally, we obtain the Fourier cosine expansion:

$$\begin{aligned} x &= \frac{l}{2} - \frac{4l}{\pi^2}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2}\cos\frac{(2k+1)\pi x}{l} \\ &= \frac{l}{2} - \frac{4l}{\pi^2}\left(\cos\frac{\pi x}{l} + \frac{1}{9}\cos\frac{3\pi x}{l} + \frac{1}{25}\cos\frac{5\pi x}{l}\cdots + \cdots\right), \quad x \in (0, l). \end{aligned} \tag{33}$$

This completes the solution.

§5.1 The coefficients: Fourier cosine series

We plot the function $\phi(x) = x$ in $0 < x < l = 3$, and partial sums $S_m(x)$.

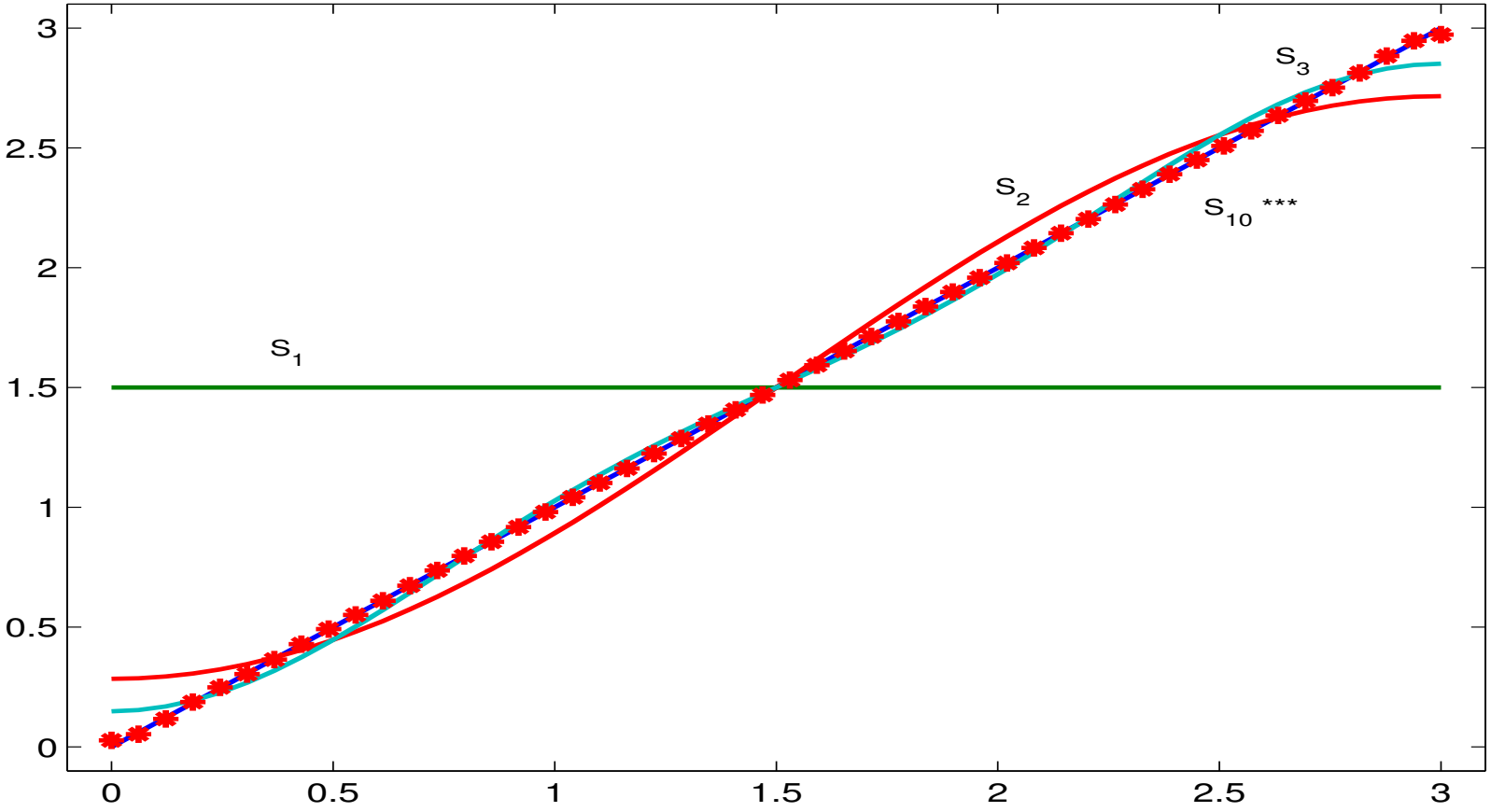

Figure: Fourier cosine series of x and S_1, S_2, S_3, S_{10} with S_{10} marked by "$*$".

§5.1 The coefficients: Full Fourier series

We have the orthogonality:

$$\int_{-l}^{l} \cos\frac{n\pi x}{l}\sin\frac{m\pi x}{l}\,dx = 0, \quad \text{for all } m, n; \qquad m, n \geq 0$$

$$\int_{-l}^{l} \cos\frac{n\pi x}{l}\cos\frac{m\pi x}{l}\,dx = 0, \quad \text{for all } m \neq n;$$

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In particular, we have

$$\int_{-l}^{l} 1\cdot\cos\frac{n\pi x}{l}dx = 0, \ \text{ for all } \ n \neq 0; \qquad \int_{-l}^{l} 1\cdot\sin\frac{n\pi x}{l}dx = 0, \ \text{ for all } \ n.$$

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In particular, we have $\left\{ \cos\frac{n\pi x}{L}, \ \sin\frac{m\pi x}{L} \right\}$

$$\int_{-l}^{l} 1 \cdot \cos\frac{n\pi x}{l}dx = 0, \quad \text{for all } n \neq 0; \qquad \int_{-l}^{l} 1 \cdot \sin\frac{n\pi x}{l}dx = 0, \quad \text{for all } n.$$

Moreover, we have $n = m > 0$ $\qquad$ $n = m = 0$

$$\int_{-l}^{l} \cos^2\frac{n\pi x}{l}dx = l = \int_{-l}^{l} \sin^2\frac{n\pi x}{l}dx, \quad n \geq 1; \qquad \int_{-l}^{l} 1dx = 2l.$$